THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET 09

Full Syllabus (Chapters 1–13) | Class 10 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** rancidity	They prevent oxidation
2	**(a)** methanoic acid	Formic acid
3	**(b)** base	Natural indicator
4	**(a)** chromium and nickel	Resists rusting
5	**(a)** Ag_2S	Silver sulphide
6	**(a)** Student A is correct	Tail is hydrophobic
7	**(c)** each C is bonded to four others in a 3-D network	Rigid structure
8	**(a)** lipase	Pancreatic/intestinal
9	**(a)** pituitary	Growth hormone release
10	**(a)** Correct	Hormonal method
11	**(a)** Both are correct	Both functions
12	**(b)** 4	RY, Ry, rY, ry
13	**(a)** males only	XY
14	**(c)** $h'/h = v/u$	Lens
15	**(c)** eye lens becomes milky and cloudy	Surgery restores vision
16	**(b)** double	Ohm's law
17	**(c)** 6 Ω	$150/25$
18	**(c)** the direction of current is reversed	Also if field reversed
19	**(b)** 1% of the sunlight falling on leaves	Photosynthesis
20	**(c)** tertiary/secondary consumer (carnivore)	Eats herbivore

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	Slow eating away of a metal by air, moisture or chemicals around it; e.g. rusting of iron (also black coating on silver, green coating on copper).	1+1
22	The process is highly exothermic; adding water to acid can make the mixture splash and the glass may break. Adding acid slowly to water keeps it safe.	2
23	Xylem (water and minerals) and phloem (food)	1+1
24	Iodine is needed for thyroxine; deficiency causes goitre.	2
25	It passes on the body design and information from parents to offspring.	2
26	Alleles; they separate so each gamete gets only one.	2
27	(i) angle of incidence = angle of reflection (ii) the incident ray, the reflected ray and the normal lie in the same plane.	1+1
28	It is a small magnet; it experiences a force due to the magnetic field of the bar magnet.	2

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	(i) displacement — Cl displaces I (ii) decomposition (iii) combination	1+1+1
30	Strong electrostatic forces between ions need much energy to overcome; NaCl (1074 K), $MgCl_2$ (981 K)	1+2
31	Secreted in small amounts, carried by blood to target organs; growth hormone controls growth; insulin controls blood sugar	1+2
32	Creates variation; helps survival in changing environments; combination of DNA from two parents leads to evolution	3
33	Ciliary muscles relax for distant objects (lens thin, long focal length) and contract for near objects (lens thick, short focal length)	3

34	It experiences a force perpendicular to both field and velocity, so its path bends (circular); Fleming's left-hand rule with current opposite to electron motion	3
35	Scale 0–14 to measure strength of acids/bases; higher H^+ concentration → lower pH; quicklime neutralises acidic soil.	1+1+1
	OR: A — HCl is a strong acid with more H^+ ions than acetic acid (weak acid).	1+2
36	Only about 10% of energy passes to the next trophic level; e.g. grass (1000 J) → deer (100 J) → lion (10 J)	1+2
	OR: Grass 10 000 J → rabbit 1000 J → fox 100 J; 90% lost as heat and used in life processes at each level	3

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(ii) $\frac{1}{v} = -\frac{1}{10} + \frac{1}{25} = -\frac{3}{50}$ ⇒ $v = -16.7$ cm; $m = -v/u = -0.67$ ⇒ $h' = -3.3$ cm (iii) real, inverted, diminished, between F and C	1+3+1
	OR: (i) converging (convex) (ii) $f = 1/2 = 0.5$ m = 50 cm (iii) $\frac{1}{v} = \frac{1}{50} - \frac{1}{25} = -\frac{1}{50}$ ⇒ $v = -50$ cm (iv) virtual, erect, magnified ($m = 2$)	1+1+2+1
38	(i) formed by sharing electrons (ii) no ions; weak intermolecular forces (iii) O shares one pair with each H with two lone pairs; N shares with 3 H with one lone pair	1+2+2
	OR: (i) soaps: Na/K salts of long-chain carboxylic acids, form scum in hard water, biodegradable; detergents: ammonium/sulphonate salts, work in hard water, often non-biodegradable (ii) non-biodegradable ones persist (iii) water containing Ca^{2+}/Mg^{2+} salts	3+1+1
39	(i) $H = I^2Rt$ (ii) electric iron, toaster, bulb, fuse (iii) $P = VI = 1100$ W; $E = 1.1 \times 2 = 2.2$ kWh	2+1+2
	OR: (i) same current; one fails — all stop; voltage divides (ii) $\frac{1}{R} = \frac{1}{4} + \frac{1}{8} + \frac{1}{8} = \frac{1}{2}$ ⇒ $R = 2\ \Omega$; $I = 4$ A; currents 2 A, 1 A, 1 A	2+3
40	(i) glucose → pyruvate (cytoplasm); without O_2: → ethanol + CO_2 (yeast); lack of O_2: → lactic acid (muscles); with O_2: → CO_2 + H_2O + energy (mitochondria) (ii) cytoplasm, mitochondria	4+1
	OR: (i) lymph is colourless, has less protein, no RBCs (ii) carries digested fats from intestine; drains excess fluid back to blood; immunity (iii) plasma, RBC, WBC, platelets (iv) blood clotting	1+2+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.